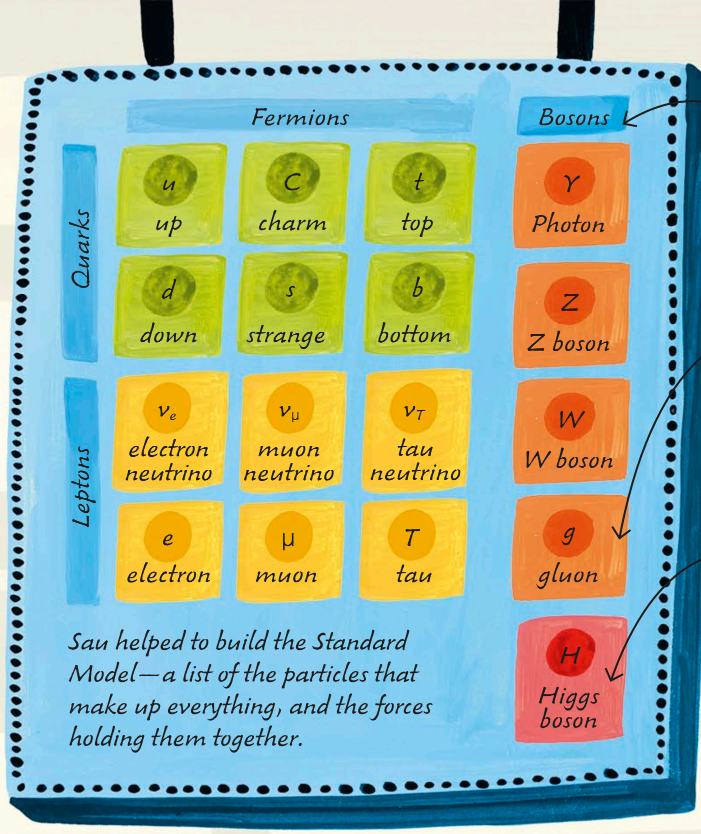
The smallest things

Sau wanted to find out more about the building blocks of the universe. At first, atoms were thought to be the particles that made up everything. Then scientists discovered atoms were made up of smaller particles, called protons, neutrons, and electrons. By the time Sau began her research, it was becoming clear that there were even smaller particles inside these. Sau helped to discover two of these—the J/psi particle and the gluon.



Sau helped to build the Standard Model—a list of the particles that make up everything, and the forces holding them together.

There are different types of particle—leptons, quarks, fermions, and bosons

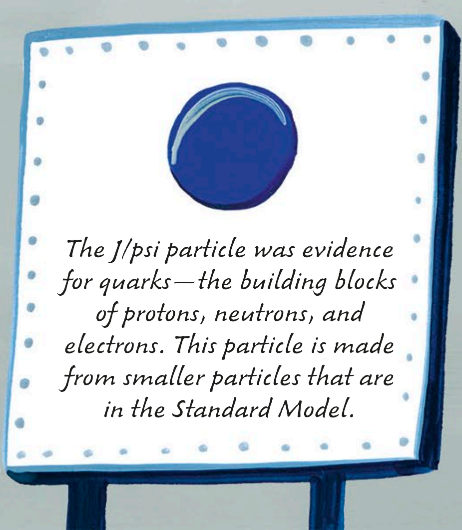
Gluons help to hold quarks together to form bigger particles.

Without the Higgs boson, other particles would zoom around all over the place at the speed of light!

... and smashed together!

The collisions produce other particles, such as Higgs bosons, but they only exist for a split second.

The Large Hadron Collider is the world's largest particle collider. It's 17 miles (27 km) long, shaped like a huge ring, and is in Geneva, Switzerland.



The J/psi particle was evidence for quarks—the building blocks of protons, neutrons, and electrons. This particle is made from smaller particles that are in the Standard Model.

Sau was also part of a huge team of scientists who spent more than 30 years hunting for the final piece of the Standard Model. In 2012, experiments using the Large Hadron Collider detected the Higgs boson, a particle like no other.

Sau is still experimenting, but she has another big mission, too. She teaches physics students how to follow in her footsteps, ready to make the world-changing discoveries of the future.